Chair of Mathematical Statistics
Department of Mathematics
Technical University of Munich

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
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Statistics for Business Administration

Exam:	MA9712 / Practice Exam C – Solutions	**Date:**	Tuesday 6th October, 2026
Examiner:	Prof. Mathias Drton	**Time:**	09:30 – 11:00

Working instructions

- This exam consists of pages with a total of 9 problems.
  Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is **52 credits**.
- Detaching pages from the exam is prohibited.
- Allowed resources:
  - one scientific pocket calculator
  - the cheat sheet, which is contained in this exam on page 16
- Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from ____________ to ____________ Early submission at _______________

Problem 1

7.5 credits

A city gym surveyed all 550 of its members about their satisfaction with a newly introduced group-class schedule. Each member answered *Positive*, *Neutral*, or *Negative*, and each member's *membership tier* (Basic, Standard, Premium) was recorded. The results are summarised in the contingency table below.

	Positive	Neutral	Negative	Total
Basic	100	90	60	250
Standard	110	55	35	200
Premium	65	20	15	100
Total	275	165	110	550

a) Identify the response variable and the explanatory variable in this study.

Solution. Response variable: satisfaction with the new schedule (Positive / Neutral / Negative). Explanatory variable: membership tier (Basic / Standard / Premium).

b) Is this study observational or experimental? Can the gym conclude that the membership tier *causes* a member's satisfaction level? Does the result generalise, and if so, to which population?

Solution. Observational — the tier was not randomly assigned. Hence no causal claim is justified; tier and satisfaction may only be associated (possible confounders). Since *all* 550 members were surveyed (a census), the figures describe exactly this gym's current membership; they generalise to that population but not necessarily to other gyms.

c) Which graphical display is appropriate to compare the distribution of satisfaction across the three tiers? Justify briefly.

Solution. A (segmented / side-by-side) bar chart of the conditional distributions of satisfaction within each tier. Both variables are categorical, so a scatterplot is inappropriate; the bar chart shows how the satisfaction split changes by tier.

d) Compute the probability that a randomly chosen member is a Premium member *and* answered Positive.

Solution. $\mathbb{P}(\text{Premium} \cap \text{Positive}) = \dfrac{65}{550} = 0.11818\ldots \approx \boxed{0.118}$.

e) Show that $\mathbb{P}(\text{Negative}) = 0.200$.

Solution. $\mathbb{P}(\text{Negative}) = \dfrac{110}{550} = 0.200.$ $\square$

f) Suppose 10 members are drawn independently at random (with replacement) from a large population in which the proportion of Negative responses equals $\mathbb{P}(\text{Negative})$ from part e). Let X be the number who answer Negative. Compute $\mathbb{P}(X = 3)$. You may use:

```
dbinom(3, size = 10, prob = 0.2)    ## [1] 0.2013266
pbinom(3, size = 10, prob = 0.2)    ## [1] 0.8791261
```

Solution. $X \sim \text{Bin}(n = 10, p = 0.2)$, so

$$\mathbb{P}(X = 3) = \binom{10}{3}(0.2)^3(0.8)^7 = 120 \cdot 0.008 \cdot 0.2097152 = 0.201327 \approx \boxed{0.201}.$$

Problem 2

6 credits

A meteorologist records the total monthly rainfall (in mm) at a weather station for a random sample of $n = 40$ months. The sample mean is $\bar{x} = 85$ mm with sample standard deviation $s = 24$ mm. A boxplot of the 40 values is roughly symmetric with no outliers; its lower and upper quartiles are $Q_1 = 68$ mm and $Q_3 = 102$ mm.

a) Read the interquartile range (IQR) of the monthly rainfall from the information given.

Solution. $\text{IQR} = Q_3 - Q_1 = 102 - 68 = \boxed{34}$ mm.

b) State the conditions required for the sampling distribution of $\bar{x}$ to be approximately normal (Central Limit Theorem), and argue whether they are met here.

Solution. (i) *Independence*: the months are a random sample and $n = 40$ is far below 10% of all possible months, so observations may be treated as independent. (ii) *Sample size / skew*: $n = 40 \geq 30$ and the boxplot is symmetric with no outliers, so the CLT applies and $\bar{x}$ is approximately normal. Both conditions are satisfied.

c) Construct the 95% confidence interval for the true mean monthly rainfall μ. You may use:

```
qnorm(0.975)    ## [1] 1.959964
```

Solution. $\bar{x} \pm z^* \dfrac{s}{\sqrt{n}} = 85 \pm 1.96 \cdot \dfrac{24}{\sqrt{40}} = 85 \pm 1.96 \cdot 3.7947 = 85 \pm 7.44.$

$$\boxed{(77.56,\ 92.44)\text{ mm.}}$$

d) Interpret your confidence interval from part c) in the context of the study.

Solution. We are 95% confident that the true mean monthly rainfall at this station is between 77.56 mm and 92.44 mm.

Problem 3

4 credits

A hardware factory produces bolts on three machines, M_1, M_2 and M_3, which account for 25%, 35% and 40% of total output, respectively. The proportion of defective bolts is 5% for M_1, 3% for M_2 and 2% for M_3. One bolt is drawn at random from the combined output.

a) Define suitable events and describe the corresponding probability tree (branches and their probabilities).

Solution. Let M_1, M_2, M_3 be "the bolt came from machine $1, 2, 3$" and D "the bolt is defective". First-level branches: $\mathbb{P}(M_1) = 0.25$, $\mathbb{P}(M_2) = 0.35$, $\mathbb{P}(M_3) = 0.40$. Second-level (conditional) branches: $\mathbb{P}(D \mid M_1) = 0.05$, $\mathbb{P}(D \mid M_2) = 0.03$, $\mathbb{P}(D \mid M_3) = 0.02$ (and the complementary $\mathbb{P}(D^c \mid M_i)$).

b) The drawn bolt is found to be defective. Compute the probability that it was produced by machine M_1. Round to three decimal places.

Solution. Law of total probability:

$$\mathbb{P}(D) = \sum_i \mathbb{P}(M_i)\mathbb{P}(D \mid M_i) = 0.25(0.05)+0.35(0.03)+0.40(0.02) = 0.0125+0.0105+0.0080 = 0.031.$$

Bayes' theorem:

$$\mathbb{P}(M_1 \mid D) = \frac{\mathbb{P}(M_1)\mathbb{P}(D \mid M_1)}{\mathbb{P}(D)} = \frac{0.0125}{0.031} = 0.40323 \approx \boxed{0.403}.$$

Problem 4

5 credits

An online retailer claims that at most 4% of all orders are returned. A consumer group inspects a random sample of $n = 200$ orders and finds that 14 of them were returned. Test, at significance level $\alpha = 0.05$, whether the data provide convincing evidence that the true return rate exceeds 4%.

a) State the null and alternative hypotheses in terms of the true return proportion p.

Solution. $H_0: \ p = 0.04 \quad$ versus $\quad H_A: \ p > 0.04$ (one-sided).

b) Let $X \sim \text{Bin}(200, 0.04)$ be the number of returned orders under H_0. Compute the p-value of the test. You may use:

```
pbinom(13, size = 200, prob = 0.04)                      ## [1] 0.9687882
pbinom(13, size = 200, prob = 0.04, lower.tail = FALSE)  ## [1] 0.0312118
```

Solution. Upper-tail (since $H_A : p > 0.04$), observed $X = 14$:

$$p\text{-value} = \mathbb{P}(X \geq 14) = 1 - \mathbb{P}(X \leq 13) = 1 - 0.9687882 = 0.031212 \approx \boxed{0.0312}.$$

c) State your decision and conclusion in context.

Solution. Since the p-value 0.0312 is less than the significance level 0.05, we reject H_0. The data provide convincing evidence that the true return rate exceeds 4%.

Problem 5

4 credits

Decide whether each of the following statements is **true** or **false** and *justify* your answer.

Remark: Only correct justifications are awarded with points, not the decision (true/false).

a) A Type I error occurs when we fail to reject a null hypothesis that is in fact false.

Solution. False. That describes a Type II error. A Type I error is *rejecting* a null hypothesis that is in fact true (a false positive).

b) If a 95% confidence interval for a parameter does not contain the value θ_0, then the two-sided test of $H_0 : \theta = \theta_0$ at level $\alpha = 0.05$ rejects H_0.

Solution. True. By the duality between two-sided tests and confidence intervals, a 95% CI is exactly the set of null values not rejected at $\alpha = 0.05$; if θ_0 lies outside the interval, the test rejects H_0.

c) Holding everything else fixed, increasing the sample size n increases the standard error of $\hat{p}$ and therefore decreases the absolute value of the test statistic $Z = (\hat{p} - p_0)/\text{SE}_0$.

Solution. False. $\text{SE}_0 = \sqrt{p_0(1 - p_0)/n}$ *decreases* as n grows, which *increases* $|Z|$ (for a fixed $\hat{p} - p_0$), making the test more powerful.

d) "We are 95% confident that μ lies in the interval" means that there is a 95% probability that this particular computed interval contains μ.

Solution. False. μ is a fixed constant and the computed interval is fixed once the data are observed, so it either contains μ or not. The 95% refers to the long-run proportion of such intervals (over repeated sampling) that would capture μ.

Problem 6

4 credits

In a clinical trial, 300 patients were randomly assigned to receive Drug A, Drug B, or a placebo (100 per group), and each patient's outcome was recorded as *Improved* or *Not improved*. The observed counts are:

	Improved	Not improved	Total
Drug A	60	40	100
Drug B	55	45	100
Placebo	35	65	100
Total	150	150	300

a) State the null and alternative hypotheses of a χ^2 test of independence between treatment group and outcome.

Solution. H_0: treatment group and outcome are independent (the improvement rate is the same across the three treatments). H_A: they are not independent (at least one treatment differs).

b) Compute the expected count in each cell and verify that the condition for the χ^2 approximation is satisfied.

Solution. $E_{ij} = \dfrac{(\text{row } i)(\text{col } j)}{300} = \dfrac{100 \cdot 150}{300} = 50$ for every cell (all row totals 100, all column totals 150). Every expected count is $50 \geq 5$, so the χ^2 approximation is valid.

c) The test statistic equals $\chi^2 = 14$. State the degrees of freedom and bracket the p-value. You may use:

```
qchisq(0.95,  df = 2)   ## [1]  5.991465
qchisq(0.99,  df = 2)   ## [1]  9.210340
qchisq(0.999, df = 2)   ## [1] 13.815511
```

Conclude in context at $\alpha = 0.05$.

Solution. $df = (r-1)(c-1) = (3-1)(2-1) = 2$. Since the observed $\chi^2 = 14 > 13.816 = q_{0.999}$, the p-value satisfies $p < 0.001$. (For reference χ^2_2 gives $p = e^{-14/2} = e^{-7} \approx 0.00091$.) Since the p-value is less than 0.05, we reject H_0. The data provide convincing evidence that the improvement rate differs across the three treatments.

Problem 7

9 credits

A retail chain tested four store layouts (W, X, Y, Z) by applying each layout to 5 randomly chosen stores and recording the average daily sales (in hundreds of euros) over one week. The group summary is:

```
# A tibble: 4 x 3
  layout      n  mean
  <chr>   <int> <dbl>
1 W           5   200
2 X           5   210
3 Y           5   190
4 Z           5   220
# grand mean = 205
```

a) State the null and alternative hypotheses of the one-way ANOVA.

Solution. $H_0: \mu_W = \mu_X = \mu_Y = \mu_Z$ (all four layouts have equal mean daily sales). H_A: at least one layout mean differs from the others.

b) State the three conditions required for ANOVA and name what you would check for each.

Solution. (i) *Independence* of observations within and across groups (random assignment of stores to layouts). (ii) *Normality* of the residuals within each group (check via normal Q–Q plot / histograms). (iii) *Constant variance* across groups (check that the group spreads are comparable, e.g. via side-by-side boxplots).

c) Complete the missing entries `(i)--(v)` in the ANOVA table below, given that the sum of squares between groups is SSG = 2500 and the sum of squares within groups (residual) is SSE = 1600.

```
             Df    Sum Sq   Mean Sq   F value
layout      (i)     2500     (iii)      (v)
Residuals   (ii)    1600     (iv)
```

Solution. With $k = 4$ groups and $n = 20$ observations:

$$\text{(i) } df_1 = k - 1 = 3, \qquad \text{(ii) } df_2 = n - k = 16.$$

$$\text{(iii) MSG} = \frac{\text{SSG}}{df_1} = \frac{2500}{3} = 833.33, \qquad \text{(iv) MSE} = \frac{\text{SSE}}{df_2} = \frac{1600}{16} = 100.$$

$$\text{(v) } F = \frac{\text{MSG}}{\text{MSE}} = \frac{833.33}{100} = \boxed{8.33}.$$

d) Carry out the F-test at $\alpha = 0.05$ and conclude in context. You may use:

```
qf(0.95, df1 = 3, df2 = 16)    ## [1] 3.238872
qf(0.99, df1 = 3, df2 = 16)    ## [1] 5.292214
```

Solution. Observed $F = 8.33$ exceeds the critical value $q_{0.95} = 3.239$ (indeed it exceeds $q_{0.99} = 5.292$), so the p-value is less than $0.01 < 0.05$. We reject H_0. The data provide convincing evidence that at least one store layout has a different mean daily sales.

e) What percentage of the total variability in daily sales is explained by the store layout?

Solution. $\text{SST} = \text{SSG} + \text{SSE} = 2500 + 1600 = 4100$, so

$$R^2 = \frac{\text{SSG}}{\text{SST}} = \frac{2500}{4100} = 0.6098 \approx \boxed{61.0\%}.$$

f) If all pairwise comparisons of the four layouts are tested afterwards using a Bonferroni correction at family-wise level $\alpha = 0.05$, what corrected significance level α^* should be used for each individual comparison?

Solution. Number of pairs $K = \binom{4}{2} = 6$, so $\alpha^* = \dfrac{\alpha}{K} = \dfrac{0.05}{6} = 0.00833 \approx \boxed{0.0083}$.

Problem 8

8.5 credits

An analyst models the fuel efficiency (`mpg`, miles per gallon) of $n = 32$ cars using their weight (`wt`, in 1000 lbs), gross horsepower (`hp`), and number of cylinders (`cyl`). The fitted `lm()` summary is:

```
Coefficients:
             Estimate  Std. Error  t value  Pr(>|t|)
(Intercept)   40.000     3.000      13.33    3.1e-13 ***
wt            -3.200     0.600      -5.33    1.2e-05 ***
hp            -0.020     0.007      -2.86    0.0080  **
cyl           -0.800     0.550      -1.45    0.1570

Residual standard error: 2.5 on 28 degrees of freedom
```

a) Write down the fitted regression equation.

Solution.

$$\widehat{\texttt{mpg}} = 40.000 - 3.200\,\texttt{wt} - 0.020\,\texttt{hp} - 0.800\,\texttt{cyl}.$$

b) Interpret the slope coefficients of `wt` and of `hp` in context.

Solution. All else equal, each additional 1000 lbs of weight is associated with a decrease of 3.20 mpg in predicted fuel efficiency. All else equal, each additional unit of horsepower is associated with a decrease of 0.020 mpg.

c) State the null and alternative hypotheses for testing whether `hp` has an effect on `mpg`, and give your decision at $\alpha = 0.05$.

Solution. $H_0: \beta_{\texttt{hp}} = 0$ versus $H_A: \beta_{\texttt{hp}} \neq 0$. The p-value is $0.0080 < 0.05$, so we reject H_0: horsepower has a statistically significant effect on `mpg`, holding `wt` and `cyl` fixed.

d) Compute the adjusted coefficient of determination R^2_{adj}. You may use:

```
var(mpg)   ## [1] 36
```

Solution. $\text{SSE} = \hat{\sigma}^2(n-k-1) = 2.5^2 \cdot 28 = 175$ and $\text{SST} = \text{Var}(\texttt{mpg})(n-1) = 36 \cdot 31 = 1116$. With $k = 3$:

$$R^2_{\text{adj}} = 1 - \frac{\text{SSE}/(n-k-1)}{\text{SST}/(n-1)} = 1 - \frac{175/28}{1116/31} = 1 - \frac{6.25}{36} = 0.82639 \approx \boxed{0.826}.$$

e) A backward elimination based on AIC is performed. Which single predictor is most likely to be dropped first, and why?

Solution. `cyl`: it has the largest p-value (0.157) and the smallest $|t|$ (1.45), i.e. it is the least significant predictor, so removing it is most likely to lower the AIC first.

Problem 9

5 credits

A university models whether an applicant is admitted (`admit` $= 1$) or rejected (`admit` $= 0$) from the applicant's grade point average (`gpa`), `gender` (`male` $= 1$, `female` $= 0$), and standardised `test_score`, including a `gpa`$\times$`gender` interaction. The fitted `glm(family = binomial)` output is:

```
Call:
glm(formula = admit ~ gpa * gender + test_score,
    family = ________ , data = applicants)

Coefficients:
              Estimate  Std. Error  z value  Pr(>|z|)
(Intercept)    -6.0000     1.4000    -4.29   1.8e-05 ***
gpa             1.5000     0.4000     3.75   0.0002  ***
gender         -2.0000     0.9000    -2.22   0.0264  *
test_score      0.0100     0.0030     3.33   0.0009  ***
gpa:gender      0.8000     0.3500     2.29   0.0221  *
```

a) Fill in the blank in the `glm()` call above (the argument to `family`).

Solution. `family = binomial` (equivalently `binomial(link = "logit")`).

b) Write the fitted model (the estimated *logit*) for a *female* applicant, simplified.

Solution. For a female, `gender` $= 0$, so the `gender` and `gpa:gender` terms vanish:

$$\widehat{\text{logit}}(p) = \log \frac{\hat{p}}{1-\hat{p}} = -6.0000 + 1.5000\,\texttt{gpa} + 0.0100\,\texttt{test\_score}.$$

c) Interpret the coefficient of `test_score` as an odds ratio.

Solution. $\text{OR} = e^{\beta_{\texttt{test\_score}}} = e^{0.0100} = 1.0101$. Each one-point increase in `test_score` multiplies the odds of admission by 1.0101; that is, the odds of admission are increased by about 1.0%, given all else being equal.

d) Using the classification table below (with a threshold of 0.5), compute the sensitivity and the specificity of the classifier. The fitted model achieves an AUC $= 0.877$; state in one sentence what this value measures.

```
                Actual admit    Actual reject
Pred. admit          40              15
Pred. reject         10              85
```

Solution. Here "admit" is the positive class: TP $= 40$, FN $= 10$, TN $= 85$, FP $= 15$.

$$\text{sensitivity} = \frac{\text{TP}}{\text{TP}+\text{FN}} = \frac{40}{50} = \boxed{0.80}, \qquad \text{specificity} = \frac{\text{TN}}{\text{TN}+\text{FP}} = \frac{85}{100} = \boxed{0.85}.$$

The AUC $= 0.877$ is the probability that the model assigns a higher predicted admission probability to a randomly chosen admitted applicant than to a randomly chosen rejected one; it measures the model's overall ability to discriminate between the two classes ($0.5 =$ chance, $1 =$ perfect).

Cheat sheet

Exploring data. $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$, $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$, $s = \sqrt{s^2}$, $\mathrm{IQR} = Q_3 - Q_1$, fences $Q_1 - 1.5\,\mathrm{IQR}$, $Q_3 + 1.5\,\mathrm{IQR}$.

$$r_{(x,y),n} = \frac{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{s_x\, s_y}$$

Probability. $\mathbb{P}(A^c) = 1-\mathbb{P}(A)$; $\mathbb{P}(A\cap B) = \mathbb{P}(A)\mathbb{P}(B)$ if independent; $\mathbb{P}(A\cup B) = \mathbb{P}(A)+\mathbb{P}(B)-\mathbb{P}(A\cap B)$; $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(A\cap B)}{\mathbb{P}(B)}$.

$$\mathbb{E}(X) = \textstyle\sum_i x_i\,\mathbb{P}(X = x_i) \ \ \text{(discrete)}, \quad \mathbb{E}(X) = \int x f(x)\,dx \ \ \text{(cont.)},$$

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2, \quad \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}, \quad \mathrm{Cov}(X,Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y), \quad Z = \frac{X - \mathbb{E}(X)}{\sqrt{\mathrm{Var}(X)}}.$$

Distributions.

$$\text{Geometric: } \mathbb{P}(X = x) = (1-p)^{x-1}p,\ \mathbb{E}(X) = \tfrac{1}{p},\ \mathrm{SD}(X) = \sqrt{\tfrac{1-p}{p^2}}.$$

$$\text{Binomial: } \mathbb{P}(X = x) = \binom{n}{x}p^x(1-p)^{n-x},\ \mathbb{E}(X) = np,\ \mathrm{SD}(X) = \sqrt{np(1-p)}.$$

$$\text{Neg. Binomial: } \mathbb{P}(X = x) = \binom{x-1}{k-1}p^k(1-p)^{x-k},\ x \ge k. \qquad \text{Poisson: } \mathbb{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!},\ \mathbb{E}(X) = \lambda,\ \mathrm{SD}(X) = \sqrt{\lambda}.$$

Inference — categorical.

$$\text{1 prop. CI: } \hat{p} \pm z^*\,\mathrm{SE}(\hat{p}),\ \mathrm{SE}(\hat{p}) = \sqrt{\tfrac{\hat{p}(1-\hat{p})}{n}}; \qquad \text{test: } Z = \frac{\hat{p} - p_0}{\mathrm{SE}_0},\ \mathrm{SE}_0 = \sqrt{\tfrac{p_0(1-p_0)}{n}}.$$

$$\text{2 prop. CI: } (\hat{p}_1 - \hat{p}_2) \pm z^*\sqrt{\tfrac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \tfrac{\hat{p}_2(1-\hat{p}_2)}{n_2}}; \quad \text{test: } \mathrm{SE}_0 = \sqrt{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}})\big(\tfrac{1}{n_1} + \tfrac{1}{n_2}\big)},\ \hat{p}_{\text{pool}} = \tfrac{\hat{p}_1 n_1 + \hat{p}_2 n_2}{n_1 + n_2}.$$

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}, \quad \text{expected}_{ij} = \tfrac{(\text{row } i)(\text{col } j)}{\text{total}}, \quad df = k-1 \text{ (GOF) or } (r-1)(c-1).$$

Inference — numerical.

$$\text{1-sample } t:\ \bar{x} \pm t^*_{df}\,\mathrm{SE}(\bar{x}),\ T_{df} = \frac{\bar{x} - \mu_0}{\mathrm{SE}(\bar{x})},\ \mathrm{SE}(\bar{x}) = \tfrac{s}{\sqrt{n}},\ df = n-1.$$

$$\text{2-sample } t:\ (\bar{x}_1 - \bar{x}_2) \pm t^*_{df}\,\mathrm{SE},\ T_{df} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\mathrm{SE}},\ \mathrm{SE} = \sqrt{\tfrac{s_1^2}{n_1} + \tfrac{s_2^2}{n_2}},\ df = \min(n_1 - 1, n_2 - 1).$$

$$\text{Sample size: two-sample } n \ge \frac{2\sigma_0^2}{\delta_0^2}(z_{1-\alpha^*} + z_{1-\beta})^2; \quad \text{one-sample } n \ge \frac{\sigma_0^2}{\delta_0^2}(z_{1-\alpha^*} + z_{1-\beta})^2,\ \alpha^* \in \{\alpha, \alpha/2\}.$$

ANOVA & regression. $F = \dfrac{\mathrm{MSG}}{\mathrm{MSE}}$, $df_1 = k-1$, $df_2 = n-k$.

$$\mathrm{SSG} = \sum_{i=1}^{k} n_i(\bar{y}_i - \bar{y})^2, \quad \mathrm{SST} = \sum_{j=1}^{n}(y_j - \bar{y})^2, \quad \mathrm{SSE} = \mathrm{SST} - \mathrm{SSG}.$$

$$\text{Post-hoc } T_{df} = \frac{\bar{x}_i - \bar{x}_j}{\sqrt{\mathrm{MSE}/n_i + \mathrm{MSE}/n_j}},\ df = n-k; \quad \text{Bonferroni } \alpha^* = \alpha/K,\ p_{\text{adj},i} = K\,p_i,\ K = \tbinom{k}{2}.$$

$$\hat{y} = b_0 + b_1 x, \quad b_1 = r_{(x,y),n}\frac{s_y}{s_x}, \quad \hat{\sigma} = \sqrt{\tfrac{1}{n-2}\textstyle\sum(y_i - \hat{y}_i)^2}, \quad T_{df} = \frac{b_1 - \beta_{1,0}}{\mathrm{SE}_{b_1}},\ df = n-2.$$

$$R^2 = \frac{\sum(\hat{y}_i - \bar{y})^2}{\sum(y_i - \bar{y})^2} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}, \quad R^2_{\text{adj}} = 1 - \frac{\sum(y_i - \hat{y}_i)^2/(n-k-1)}{\sum(y_i - \bar{y})^2/(n-1)}, \quad \mathrm{AIC} = c + n\ln\big(\textstyle\sum e_i^2\big) + 2k.$$

Logistic regression. $\mathrm{logit}(p_i) = \log\frac{p_i}{1-p_i} = \beta_0 + \beta_1 x_{1,i} + \cdots + \beta_k x_{k,i}$, $\mathrm{OR}_j = e^{\beta_j}$, sensitivity $= \mathbb{P}(\text{Test+} \mid \text{Cond+})$, specificity $= \mathbb{P}(\text{Test}- \mid \text{Cond}-)$.

Quantile hints (standard normal). $z_{0.70} = 0.524$, $z_{0.80} = 0.842$, $z_{0.90} = 1.282$, $z_{0.95} = 1.645$, $z_{0.975} = 1.960$, $z_{0.99} = 2.326$, $z_{0.995} = 2.576$.